

The longest mission

Read about the longest running mission in space ever. The story of the Voyager I probe
<http://1.usa.gov/ijdXsv>

Space menu

A series of images showcasing the various foods that the average astronaut enjoys in space
<http://bit.ly/wtbxsF>

COSMIC LOGISTICS

What will space travel be like a decade from now? If we suddenly discover life on a distant planet, how will we get there? What are the logistics involved in such an endeavour?

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Space is vast, the sheer scale of it is mind-boggling. Even within our solar system, distances have to be measured in millions of kilometres, anything smaller would just have too many zeroes in it. It isn't just the distance, the speeds you have to deal with are also incredibly large. The voyager space probe for example, travels at over 17km/s, which is still only a fraction of the speed of the Earth around the sun. This just a tenth of the speed at which the sun moves around the central axis of our galaxy. The problem is that even at that speed, it would still take us over 50 years just to get to the edge of the solar system; 70,000 to reach

our nearest star. How does one plan for such a trip? The spacecraft can manage fine on its own, but what about us humans? What will we eat? Can we even survive such a trip? Read on to find out.

The problem

At present, dining in space is almost as good as enjoying a meal on Earth, once scientists discovered that all they had to do was make the food sticky and reduce the portion sizes, astronauts are being spoiled silly by the choice of food available. The Russians alone have a menu sporting over 300 items and enjoy everything from soup to mashed potatoes and roasted meat. The shelf life of the food products, depending on their type and how they are stored, ranges from a few months to many years.

The problem therefore is not what to eat, but rather where and how to store that food. An average meal packet is about 16 cubic inches in volume, an astronaut will



Astronauts can have almost all their favourite foods in space

need at least four such packets per meal, three meals a day, resulting in a little over a cubic metre of storage per year per astronaut. This does not even include the water that would be required, for drinking as well as re-hydration of the food. One cubic metre of food would weigh more than a tonne on the Earth and if you consider a trip lasting over 50 years, that would be more than 50 cubic metres of volume just for food alone. 200 tonnes of food for a crew of 4 onboard and as many cubic metres of storage.

